



03/21/2024

Certificate of Tests

PAGE: 6

Purchase Order: 26604262

Sales Order: 1001000510

Heat Number

Vendor Name

Mfg. Date

item# 1233153 **2237328**

Nucor Hickman - Armored AR US

03/04/2024

Item Number

Description

11000102

5L- X52 4-1/2X.337 (15.00#)BR,DRL,PEB

CHEMICAL ANALYSIS

	C	Mn	P	S	Ni	Cr	Cu	Mo	Si	Ti	V	CE
Ladle	0.0500	1.0700	0.0110	0.0012	0.0400	0.0500	0.1200	0.0100	0.2100	0.0140	0.0040	0.1208
	Al:	0.0400	Sn:	0.0050	N:	0.0060	B:	0.0000	Ca:	0.0020	Nb:	0.0310
Chem	0.0630	1.1163	0.0086	0.0009	0.0443	0.0410	0.1094	0.0100	0.2163	0.0126	0.0042	0.1360
	Al:	0.0357	Sn:	0.0045	N:	0.0201	B:	0.0001	Ca:	0.0023	Nb:	0.0323
Chem	0.0730	1.1130	0.0082	0.0007	0.0437	0.0420	0.1062	0.0096	0.2163	0.0125	0.0043	0.1457
	Al:	0.0355	Sn:	0.0044	N:	0.0358	B:	0.0001	Ca:	0.0022	Nb:	0.0319

Acceptance Criteria: CEPCM = 0.2500

MECHANICAL PROPERTIES

YS (ksi): 70.01 TS (ksi): 74.10

EL (%) : 29.00 HRB: 88.00 Y/T(%): 0.94

CHARPY TEST RESULTS BODY

ORIENTATION:	TRANSVERSE	Specimen(S)	Energy (ft/lbs.)	%Shear
Spec. Size (in) :	.394"X.197"X2.165"	1	80.00	100.00
Test Temp,(of)	-20	2	78.00	100.00
Location	3T90	3	77.00	100.00
		Average	78.33	100.00

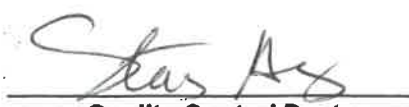
Acceptance Criteria Min. FT/lbs: Min. Ft/Lbs 10

COMMENTS

MINIMUM HYDROSTATIC TEST PRESSURE (PSI) : 4,700 PER 5 SECONDS

MANUFACTURED TO THE 46TH EDITION OF API 5L SPECIFICATION
MANUFACTURED IN THE USA-HIGH FREQUENCY WELDED PRODUCT
WELD SEAM ANNEALING AT 1600 DEGREES F.MINIMUM
ULTRASONIC TESTED TO N-10 NOTCH 1/8 DRILLED HOLE
FLATTENING TEST PASSED
1 INCH L90 STRIP SPECIMEN
THIS MATERIAL MEETS :ASTM A53B-22,ASME SA53B-21 LATEST EDITION
API 5L PSL 2 GRADE X52M
MEETS HARDNESS REQUIREMENTS OF NACE MRO 175/ISO 15156-2, CLAUSE A.2.1.1
MEETS ISO 10474 3.1.B/EN 10204: 3.1 / STANDARDS ARE LATEST REVISION
MELTED AND MANUFACTURED IN THE U.S.A.
STEEL SUPPLIER: Nucor Hickman - Armored AR US
COIL MANUFACTURER: Nucor Hickman - Armored AR US

The above analyses and tensile properties are correct to the best of my knowledge and belief.


Quality Control Dept